

# COMBINATORICS

## NOTES JANUARY 29, 2026

### 1. STERLING NUMBERS. CON'T

Let  $f : M \rightarrow B$  be a function where  $M$  is a set of marbles and  $B$  is a set of boxes. Also let  $m = |M|$  and  $b = |B|$ . We concern ourselves with the following counts.

- (1) We want the count of the number of relations from  $M$  to  $B$ :

$$\text{count} = 2^{|M \times B|} = 2^{|M||B|} = 2^{mb}$$

- (2) We want the count of the number of general functions from  $M$  to  $B$  which follows as:

$$\text{count} = b^m$$

- (3) With  $m \leq b$  we want to count the number of one-to-one functions from  $M$  to  $B$  which follows as:

$$\text{count} = {}_bP_m$$

- (4) With  $m \leq b$  we want to count the number of onto functions from  $M$  to  $B$ , which follows as:

$$\begin{aligned} \text{count} &= b^m - \binom{b}{1}(b-1)^m + \binom{b}{2}(b-2)^m - \binom{b}{3}(b-3)^m + \cdots + (-1)^b \binom{b}{b}(b-b)^m \\ &= b! \cdot \frac{1}{b!} \sum_{k=0}^b (-1)^k \binom{b}{k} (b-k)^m = b!S(m, b) \end{aligned}$$

- (5) With  $m = b$  we want to count the number of bijections from  $M$  to  $B$ . This follows as

$$\text{count} = m! = b!$$

Now we consider the diagram

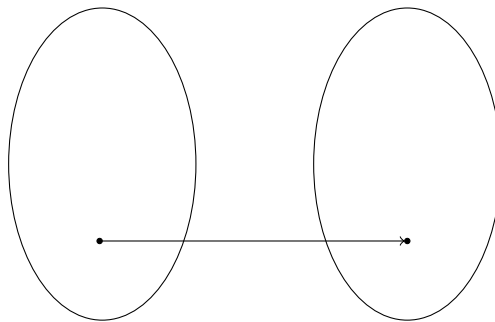


FIGURE 1. Set mapping

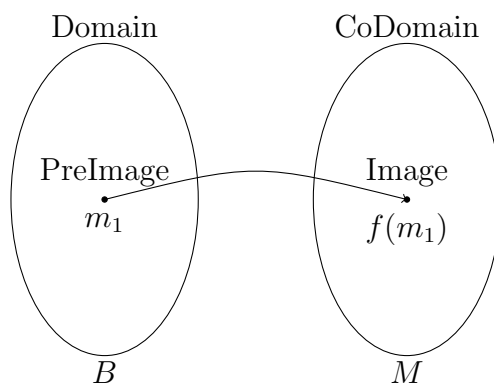


FIGURE 2. Set mapping

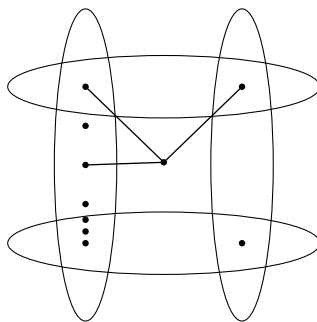


FIGURE 3. Something here

Linear Functions

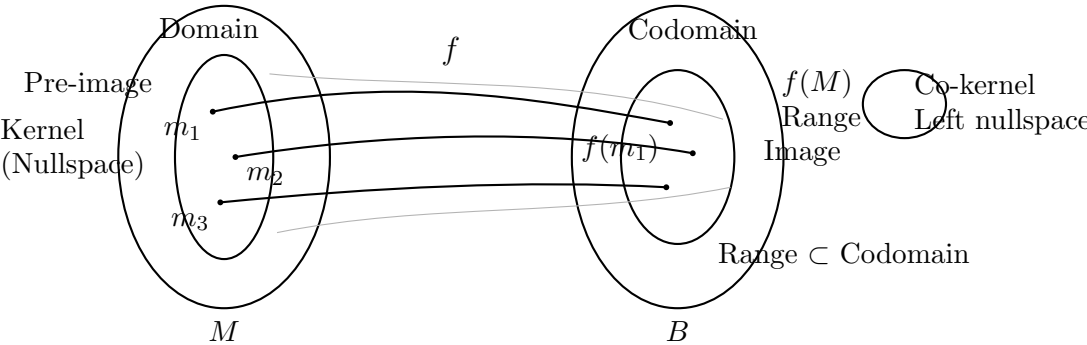


FIGURE 4. Chat GPT Tikz

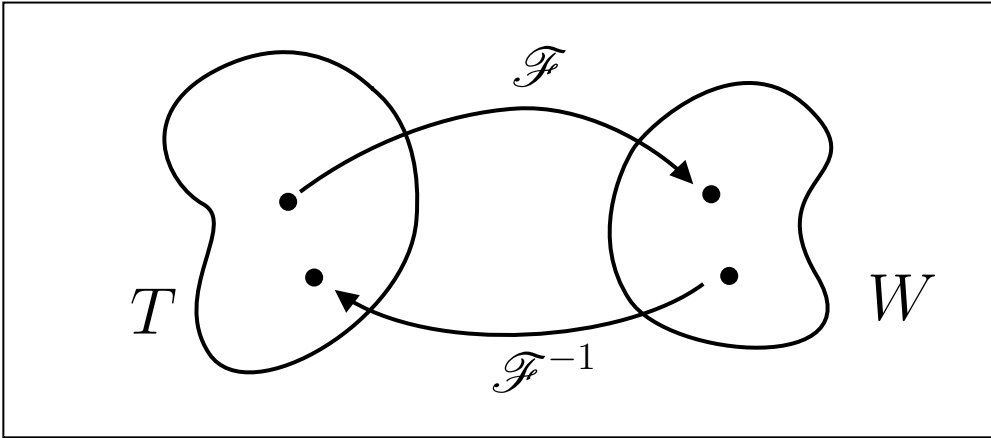


FIGURE 5. Something here